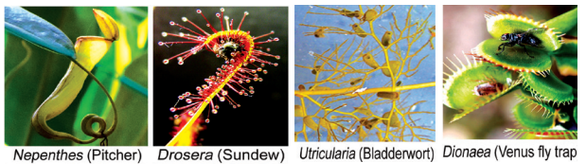
lid. When insect is trapped, proteolytic enzymes will digest the insect.

b. ***Drosera*** (Sundew): It consists of long club shaped leaves with tentacles that secrete sticky digestive fluid which looks like a sundew and attracts insects.

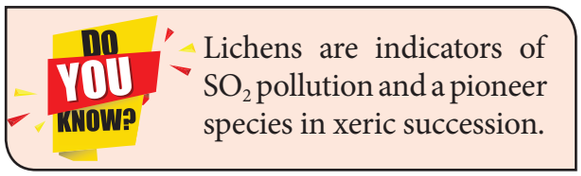
c. ***Utricularia*** (Bladder wort): Submerged plant in which leaf is modified into a bladder to collect insect in water.

d. ***Dionaea*** (Venus fly trap): Leaf of this plant modified into a colourful trap. Two folds of lamina consist of sensitive trigger hairs and when insects touch the hairs it will close and traps the insects.(Figure 12.12).



Nepenthes (Pitcher) *Drosera* (Sundew) *Utricularia* (Bladderwort) *Dionaea* (Venus fly trap

Figure 12.12: Insectivorous mode of nutrition



Lichens are indicators of SO_2 pollution and a pioneer species in xeric succession.

Check your grasp!

Mineral X required for the activation of enzyme nitrogenase, Mineral Y involved in transport of sugar and Mineral Z required for maintaining ribosome structure. Identify X, Y and Z.

Summary

Sources of minerals for plants are atmosphere, water and soil. Minerals are classified based on their quantity, mobility and functions. Macro nutrients (C, H, O, N, P, K, Ca, Mg and S) are required in higher concentration and micro nutrients (Fe, Mn, Cu, Zn, B, Mo, Cl and Ni) are required in lesser concentration. Minerals like Sodium, Cobalt, Silicon and Selenium are required by some plants for specific functions and such minerals are grouped as unclassified minerals. Actively mobile elements are N, P, K, Mg, Cl, Na, Zn and Mo. The deficiency symptoms for these minerals first appear on old and senescent leaves due to active movement of minerals to younger leaves. Relatively immobile elements are Ca, S, Fe, B and Cu. In such minerals, deficiency symptoms first appear on young leaves due to immobile nature. Minerals and their deficiency symptoms include chlorosis (loss of chlorophyll pigments), necrosis (death of tissue), anthocyanin formation, die back of shoot, exanthema, hooked leaf tip, whiptail and so on. A concentration at which 10% of dry weight is reduced is considered as critical concentration. Minerals used in excess concentration become toxic.

Soil less cultivation alleviates problems due to mineral deficiency. It includes hydroponics and aeroponics. Hydroponics is a method of growing plants in a nutrient solution. Aeroponics is the technique in which roots are suspended in air and nutrient sprayed over the roots by motor driven rotor. Nitrogen is an important requirement for normal growth and functioning of a plant. Nitrogen fixing organisms fix nitrogen from atmosphere naturally through symbiotic and non-symbiotic modes. Special modes of nutrition are seen in plant which grew in nutrient deficient soils and the character becomes permanent.

Evaluation

1. Identify correct match.

   1. Die back disease of citrus - (i) Mo
   2. Whip tail disease - (ii) Zn
   3. Brown heart of turnip - (iii) Cu
   4. Little leaf - (iv) B

   a. 1 (iii) 2 (ii) 3 (iv) 4 (i)
   b. 1 (iii) 2 (i) 3 (iv) 4 (ii)
   c. 1 (i) 2 (iii) 3 (ii) 4 (iv)
   d. 1 (iii) 2 (iv) 3 (ii) 4 (i)

2. If a plant is provided with all mineral nutrients but, Mn concentration is increased, what will be the deficiency?

   a. Mn prevent the uptake of Fe, Mg but not Ca
   b. Mn increase the uptake of Fe, Mg and Ca
   c. Only increase the uptake of Ca
   d. Prevent the uptake Fe, Mg, and Ca



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